

NanoGT Match Report: Kohlschutter-Tonz syndrome

Disease: Kohlschutter-Tonz syndrome (ORPHA:1946)

Primary gene: ROGDI

Gene CDS: 861 bp

Inheritance: Autosomal recessive

Target tissues: CNS

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	ABO-101	AAV9	8.5/10	High	phase1/2
2	RGX-121	AAV9	8.5/10	High	phase3
3	OAV101-IT	AAV9	8.0/10	High	approved
4	Zolgensma	AAV9	8.0/10	High	approved
5	DTX201	AAV8	7.3/10	Medium	phase2

Match #1: ABO-101

Precedent disease: Mucopolysaccharidosis type IIIB

Vector: AAV9

Tissue target: CNS

Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	0.5	1.0

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Both secreted proteins — systemic delivery viable
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: phase1/2

Match #2: RGX-121

Precedent disease: Mucopolysaccharidosis type II

Vector: AAV9

Tissue target: CNS/liver

Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	2.0	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.8	1.0

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Both secreted proteins — systemic delivery viable
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: phase3

Match #3: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.0	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Partial match: extracellular/secreted component present
- Inheritance match (Autosomal recessive <-> AR)

- Unknown pathway — neutral score
- Approval status: approved

Match #4: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 8.0 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	2.0	2.0
Protein class	1.0	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	1.0	1.0
Approval precedent	1.0	1.0

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)
- Vector tropism plus precedent target match: cns
- Partial match: extracellular/secreted component present
- Inheritance match (Autosomal recessive <-> AR)
- Unknown pathway — neutral score
- Approval status: approved

Match #5: DTX201

Precedent disease: Hemophilia A

Vector: AAV8

Tissue target: liver

Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max
Packaging fit	2.0	2.0
Tissue tropism	1.0	2.0
Protein class	2.0	2.0
Pathway similarity	1.0	2.0
Inheritance compatibility	0.7	1.0
Approval precedent	0.6	1.0

Rationale

- Gene CDS 861bp / cargo 4700bp (18% utilized)

- Vector tropism overlaps cns, but precedent target is liver
- Both secreted proteins — systemic delivery viable
- LOF inheritance — compatible for gene replacement
- Unknown pathway — neutral score
- Approval status: phase2